

SUYASH FULAY

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INTERESTS

I study large-scale populations computationally using language and graph models, probe how well these models align with different groups, and build systems that leverage AI to improve collective decision-making. I've approached these problems as both a researcher and full-stack engineer, with experience in industry and academia.

EDUCATION

- 2022–2025 **Massachusetts Institute of Technology** – *Ph.D., MIT Media Lab (GPA: 5.0/5.0)*
Thesis: Language Models to Understand, Reflect, and Address Social Fragmentation
Committee: Deb Roy (advisor), Michiel Bakker & Diyi Yang
- 2020–2021 **Massachusetts Institute of Technology** – *M.Eng. in Computer Science, Concentration in Artificial Intelligence (GPA: 5.0/5.0)*
- 2013–2017 **Massachusetts Institute of Technology** – *B.S. in Computer Science and Engineering*

EXPERIENCE

- 2022–2025 **MIT Media Lab** – *Ph.D. Researcher*
Research spanning NLP, AI alignment, network science, and computational social science.
- **AI for collective decision-making:** Built AI voice agents that collected and analyzed policy-preference data at scale, gathering hundreds of hours of in-depth interviews in days rather than months [3, 4]; deployed agents that listen and respond to people in live deliberations to introduce different perspectives [6]; and developed tools enabling policymakers to elicit higher-quality preference data from constituents [7].
 - **LLM methods and alignment:** Co-developed ConGraT, self-supervised contrastive pre-training for joint graph and text embeddings [9]. Identified fundamental trade-offs in LLM alignment related to truthfulness, political bias, and AI representation [1, 8].
 - **Social network analysis:** Conducted one of the largest computational studies of language polarization on social networks (analyzing 2.5 billion comments over the past fifteen years), and proposed novel graph metrics to measure preferences on these networks [2, 10].
- JUN 2024
SEP 2024 **Amazon AGI Foundations** – *Applied Scientist Intern*
Introduced a suite of improvements to the alignment methods for Amazon's Titan model (e.g. curriculum learning, novel loss functions) that allowed the use of finer-grained human preference data, leading to improvements on internal and external benchmarks.
- JUN 2022
AUG 2022 **Mana AI** – *ML Research Intern*
Built the founder- and company-identification engine for the first iteration of Mana's AI investor, applying graph neural networks and NLP to LinkedIn and Twitter data to surface and evaluate investment opportunities autonomously.
- 2017–2020 **Bridgewater Associates** – *Researcher and Engineer*
Founding member of ML team at world's largest hedge fund. Built a machine-learning pipeline — spanning software engineering, data science, and investment research — now used by researchers to modify trading systems across a range of macroeconomic and financial indicators. Mentored by Dr. Jasjeet Sekhon, Yale/Berkeley professor of statistics and Chief Strategy Officer at Google DeepMind.

SELECTED PUBLICATIONS

- [1] From Delegates to Trustees: How Optimizing for Long-Term Interests Shapes Bias and Alignment in LLMs
Suyash Fulay, Jocelyn Zhu, Michiel Bakker. European Chapter of the Association for Computational Linguistics (**EACL**) 2026. Oral talk at International Association for Safe and Ethical AI (**IASEAI**) 2026.
- [2] Language Model Estimates Indicate Rising Outgroup Animosity Among a Sample of Partisan Users, Journalists, and Politicians on Reddit and Twitter
Suyash Fulay, Deb Roy. **Scientific Reports** 2026.
- [3] Agora: Teaching the Skill of Consensus-Finding with AI Personas Grounded in Human Voice
Suyash Fulay, Prerna Ravi, Om Gokhale, Eugene Yi, Michiel Bakker, Deb Roy. ACM Conference on Human Factors in Computing Systems, Extended Abstracts (**CHI EA**).
- [4] AI and Collective Decisions: Strengthening Legitimacy and Losers' Consent
Suyash Fulay, Prerna Ravi, Emily Kubin, Shreshtha Mohanty, Michiel Bakker, Deb Roy. ACM Symposium on User Interface Software and Technology (**UIST**) (under review).
- [5] Shareholder Democracy with AI Representatives
Suyash Fulay, Sercan Demir, Galen Hines-Pierce, Hélène Landemore, Michiel Bakker. **arXiv** preprint 2025.
- [6] Using LLMs to Raise Missing Perspectives in Policy Deliberations
Suyash Fulay, Dimitra Dimitrakopoulou, Deb Roy. Neural Information Processing Systems (**NeurIPS**) 2025 Workshop on Persona Modeling, Oral talk (top 10%).
- [7] Coalesce: An Accessible Mixed-Initiative System for Designing Community-Centric Questionnaires
Cassandra Overney, Daniel Kessler, **Suyash Fulay**, Mahmood Jasin, Deb Roy. ACM Conference on Intelligent User Interfaces (**IUI**) 2025.
- [8] On the Relationship between Truth and Political Bias in Language Models
Suyash Fulay, William Brannon, Shreshtha Mohanty, Cassandra Overney, Elinor Poole-Dayana, Deb Roy, Jad Kabbara. Empirical Methods in Natural Language Processing (**EMNLP**) 2024.
- [9] ConGraT: Self-Supervised Contrastive Pretraining for Joint Graph and Text Embeddings
Will Brannon, Wonjune Kang, **Suyash Fulay**, Hang Jiang, Brandon Roy, Jad Kabbara, Deb Roy. Association for Computational Linguistics (**ACL**) 2024 TextGraphs Workshop.
- [10] Divergences in Following Patterns between Influential Twitter Users and Their Audiences
Suyash Fulay, Nabeel Gillani, Deb Roy. International AAAI Conference on Web and Social Media (**ICWSM**) 2023.

SERVICE & TEACHING

2026	Emerson Consequential Scholar One of twenty MIT PhD students selected for a program supporting the translation of research into societally impactful ventures.
2022 PRESENT	Research Mentor • Jocelyn Zhu, MIT, M.Eng. in Computer Science (2024–2025) • Ajay Arora, MIT, M.Eng. in Computer Science (2024–2025) • Isabella Quan, MIT Undergrad, Computer Science (2023–2024) • Hannah Kimura, Wellesley, Computer Science (2023–2024)
2023	Instructor, Gen AI for Constructive Communication Taught ~50 students on large language models, covering LLM fundamentals and methods for experimentation and evaluation.